

Offline metrics: exact match, recall, and the BLEU trap

recall@k over a fixed retrieval

```
relevant (ground truth): ['latte-doc', 'milk-doc', 'oat-doc']  
retrieved, top 3:        ['latte-doc', 'unrelated-doc', 'milk-doc']  
hit:                     ['latte-doc', 'milk-doc']
```

```
recall@3 = 2/3 = 0.667
```

A retrieval quality metric: of what actually mattered, how much did the top-k surface.